

NEHAL ABDUL REHMAN

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Open to Relocation

SKILLS

Technical:

- C++ | Embedded C | OOP | Python | Embedded Linux | RTOS | JTAG - JLink | OpenWRT | WSL Buildroot Yocto | OTA
- BLE WIFI/ESPNOV Z-Wave | MQTT(s) WebSocket HTTP(s) | I2C UART SPI Modbus | QT LVGL SquareLine |
- ESP IDF | STM32cube | Keil uVision | MPLAB | Segger ARM | Simplicity Studio | CC Studio | Atmel S7 | NVIDIA
- Git VCS | GitHub Actions | GitLab CI/CD | Tags/Releases | GitHub Projects & Issues | Branching & Merging & SCM
- AI Prompting | OpenCV | Documentation & Flow Diagrams | LabVIEW | MATLAB | Altium | msWord Excel Sheets Docs |

Interpersonal: Collaboration & communication | Ownership & initiative | Problem Solving & Adaptability | Technical learner

PROFESSIONAL EXPERIENCE

Poultia, Inc; Embedded IoT Engineer | Jan 2026 - Present

Key Responsibilities:

- **Embedded Software Development:** Develop memory-efficient, modular, and scalable embedded software with RTOS in C/C++ for edge IoT Node Device based on ESP32 and ESP-IDF, Complete pipeline consists of Over-The-Air (OTA) updates, CI with unit testing and linting to CD with Hardware-In-the-Loop (HIL). Along with Python and Bash scripting for automation and testing.
- **Documentation and Collaboration:** Creating and maintaining comprehensive documentation, including UML and sequential-flow diagrams and collaborating with cross-functional teams to ensure alignment with project goals.
- **Agile & Scrum Methodologies:** Actively engaging in Scrum and Agile practices, contributing to sprint planning and utilizing TeamWork & Jira project management tools for tracking project milestones.
- **Version Control System (VCS):** Source Code Management (SCM) using Git, ensuring code integrity and collaboration. Along with Continuous Integration and Continuous Delivery (GitHub actions, CI/CD).

eSpark Consulting Group; Embedded Firmware Engineer | 2024 – 2026

Achievements:

- **Engineered multi-platform embedded systems** like STM32, TI C2000, ESP32 and Embedded Linux (RK3566, Allwinner V3s, NXP iMX93), delivering end-to-end solutions including a **multi-channel smart meter**, and **industrial motor controller**.
- **Delivered embedded Linux solutions** by customizing Buildroot-based SDKs, enabling **secure IoT communication (MQTTS, SFTP)** and **hardware-accelerated video streaming (GStreamer, H.264)** with optimized performance and low latency.
- **Developed advanced user interfaces and connectivity features**, including **Qt-based GUIs for Linux systems** and **LVGL-driven embedded displays**, along with **Bluetooth A2DP audio streaming** integration on ESP32 platforms.
- **Accelerated debugging and system stability** by performing deep fault analysis (memory leaks, corruption, pointer tracing) using J-Link and XDS100, reducing system crashes and improving firmware robustness across multi-threaded environments.

UTF – LABS; Junior Embedded Firmware Developer | 2022 – 2024.

Achievements:

- **Optimized RTOS-based firmware performance and stability** by reducing memory footprint, eliminating race conditions, and implementing efficient stack/heap utilization, preventing memory leaks and improving system reliability in multi-threaded environments.
- **Customized and integrated low-level embedded drivers** for components including **3D IMU sensors, power management ICs, and IPS LCD displays**, achieving precise hardware control and meeting project-specific performance requirements.
- **Implemented advanced BLE 5.0 communication systems**, enabling **dual-mode operation (central + peripheral)** and high-speed transmission of large log datasets using **2M PHY**, significantly improving data throughput and efficiency.
- **Automated development and deployment workflows** using **bash scripting**, enabling one-click environment setup (Python, libraries, VLC) and orchestrating multi-process systems including **Node.js-based UI with WebSocket integration** for voice assistant applications across Linux and Windows.

EDUCATION

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY (NEDUET);

GPA: 3.58/4.0

Bachelor's in Industrial Electronics Engineering | Ranked 3rd in department

Concentrations: Embedded Systems, Processes Control & Automation, Power Electronics

Paper: [Adaptive Gaussian & Double Thresholding for Contour Detection and Character Recognition of 2D Area Using Computer Vision](#)

FEATURED PROJECTS

NVIDIA ORIN BASED CAMERA DRIVER DEVELOPEMENT | Embedded Linux (Linux For Tegra - L4T)

- Developed multi MIPI-CSI lane and resolution configuration driver development for AP1302 camera in C and C++.
- Modified device tree files for multiple camera integration and scripted python scripts for automation and config injection.

ALLWINNER V3S-BASED MULTI-CHANNEL SMART METER | Embedded Linux (Linaro Toolchain + Buildroot)

- Developed a smart metering system on a custom **Allwinner V3s SoC** board with multi-channel data acquisition and processing.
- Designed a responsive **QT-based GUI** integrated with **MQTTs (AWS IoT Core)** and **SFTP** for secure data exchange.

RK3566-BASED WIRELESS MIPI-CSI VIDEO TRANSMITTER / RECEIVER | Embedded Linux (Firefly SDK + Buildroot)

- Developed a wireless video transmission system on **RK3566** using **LT7911D** for MIPI-CSI conversion and **AP6275S (5 GHz)**.
- Implemented **GStreamer pipelines** with hardware H.264 encoding/decoding for low-latency video.
- Customized **Firefly SDK + Buildroot** for device bring-up, driver integration, and seamless HDMI playback.

STM32-BASED RO PURIFICATION CONTROLLER | ARM Cortex MCU

- Developed a multi-sequence menu system for **RO purification controller** based on **STM32F0xx**.
- Implemented a memory efficient, non-blocking sequential architecture with readable C/C++ code.
- Worked on custom **SOC** board with **JLINK** debugging for memory traces and pointers references.

TEXAS INSTRUMENT BASED PERMANENT AC MOTOR CONTROLLER | ARM Cortex MCU

- Developed a motor controller sequence using **TMS320C28x** and **ESP32** for **Modbus** and **HMI**.
- Implemented over current (**OCP**) and voltage (**OVP**) protection algo along with recovery states from **EEPROM**.
- Worked on custom SOC board with XDS100 debugger on Code Composer Studio.

MEDITATION DEVICE | UI Design & BT Stream

- Designed GUI on ST7789, ST77916 based on LVGL^9.2.2 on ESP32-WROOM32E with PSRAM and LVGL^8.3 on ST7262 with ESP32S3-N16R8 QSPI.
- Implemented Bluetooth Classic A2DP for audio transfer to AirPods using SD card module on SPI.

M5STACK PROJECT X | Confidentiality Agreement in Place (NDA)

- Developed firmware for an ongoing project utilizing M5Stack modules ESP-IDF SDK.
- Implemented features using C++ and FreeRTOS, enhancing the functionality and user experience.
- Integrated MQTT communication for OTA updates and game control with a Raspberry Pi 4.

SMART CHARGER USING ESP32 | IoT Development

- Developed a BLE server for WIFI provisioning, later converted to client for custom protocol.
- Implemented data parsing and logging to EEPROMs, transmitting JSON data to a WebSocket server.
- Established efficient BLE communication for large data chunks between central and peripheral devices.

VBOX ON RASPBERRY PI 4 | Voice Assistant Development

- Engineered a GPT Turbo API based voice assistant for query processing, integrating a wake word detection mechanism for user interaction on a Raspberry Pi 4 using Python.
- Utilized WebSocket communication for interaction with the user interface (also integrated on windows).